

12 (long)

(a) For a pre-cracked specimen subjected to fatigue loading, sketch the logarithm of crack growth per cycle da/dN as a function of the logarithm of the stress intensity factor range ΔK . On your diagram, label and discuss the different regions. [6]

(b) A thin-walled, cylindrical steel pressure vessel of 8 m diameter and 40 mm wall thickness is to operate at a working pressure of 4.1 MPa. Non-destructive testing revealed the presence of small semi-circular cracks in the inside wall which will gradually extend through the wall by fatigue. Assume for this crack geometry that $K = \sigma\sqrt{\pi c}$, where c is the radius of a semi-circular crack and σ is the hoop stress in the vessel. The fracture toughness K_{IC} of the steel is $200 \text{ MPa m}^{1/2}$.

(i) Will the vessel fail in service by leaking or by fast fracture? [6]

(ii) During service, the vessel is subjected to repeated cycles from zero to working pressure. In fatigue conditions, crack growth is given by Paris' law (page 7 of the Materials Databook), with $n = 4$ and $A = 2.44 \times 10^{-14} \text{ MPa}^{-4} \text{ m}^{-1}$. Find the initial crack size c_0 if the vessel is to survive 2000 pressurisation cycles. [10]

(iii) The vessel is subjected to a proof test so that any initial cracks larger than c_0 will fracture the vessel catastrophically. Find the pressure to which the vessel must be subjected to survive the proof test. [8]

END OF PAPER